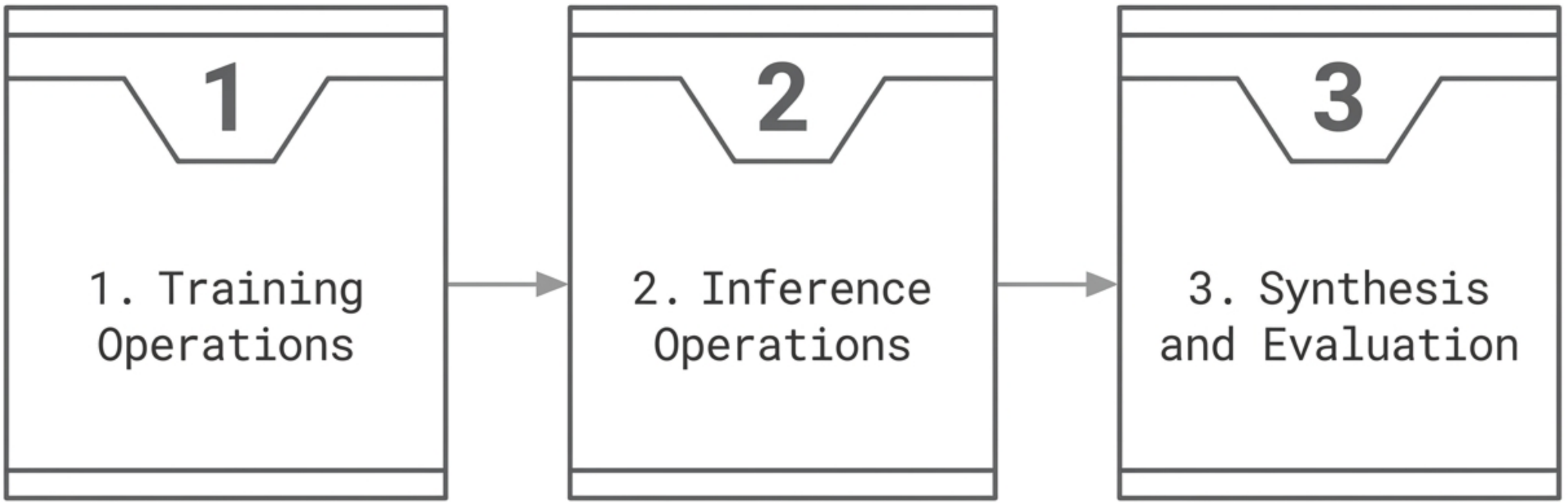
MODULE 13, LESSON 3

AI 101 FOR MECHANICAL ENGINEERS

WHAT CHANGES WEIGHTS AND WHAT DOES NOT

Gemini Notebook

Lesson Structure



Gemini Notebook

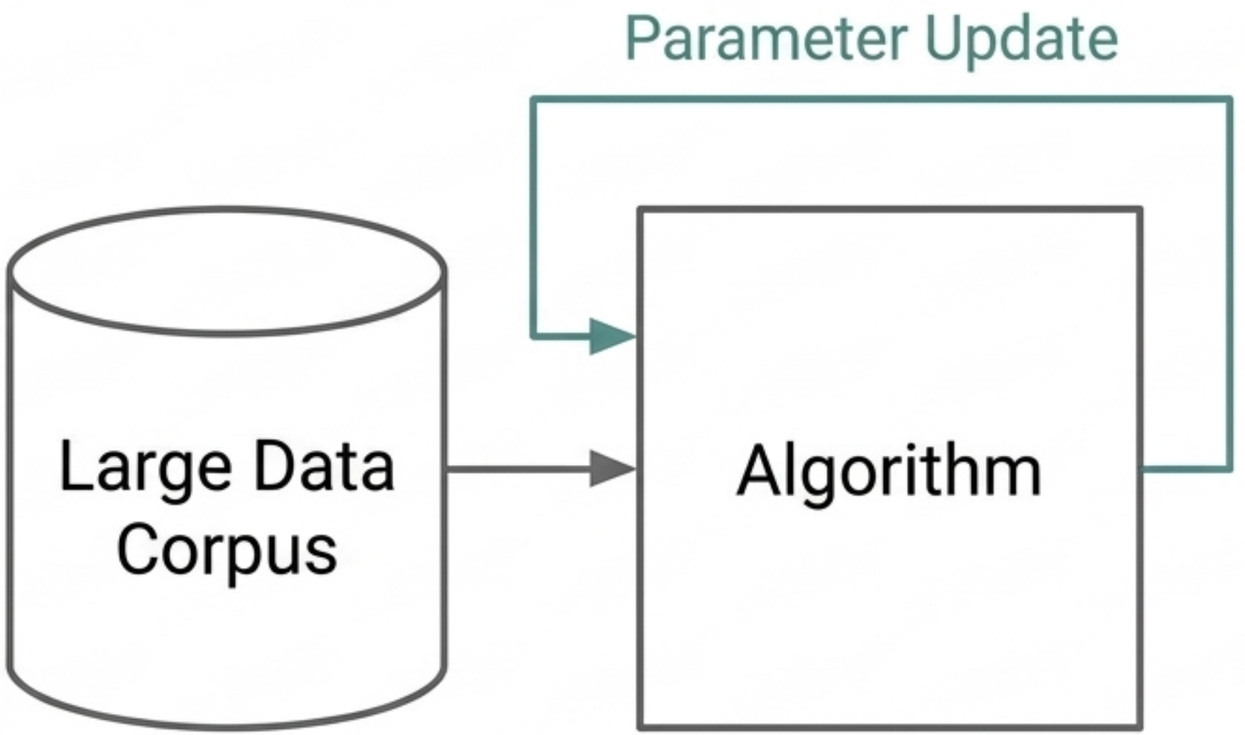
1. Training Operations

Purpose: Identify machine learning operations that update internal model parameters.

Gemini Notebook

Pretraining Mechanics

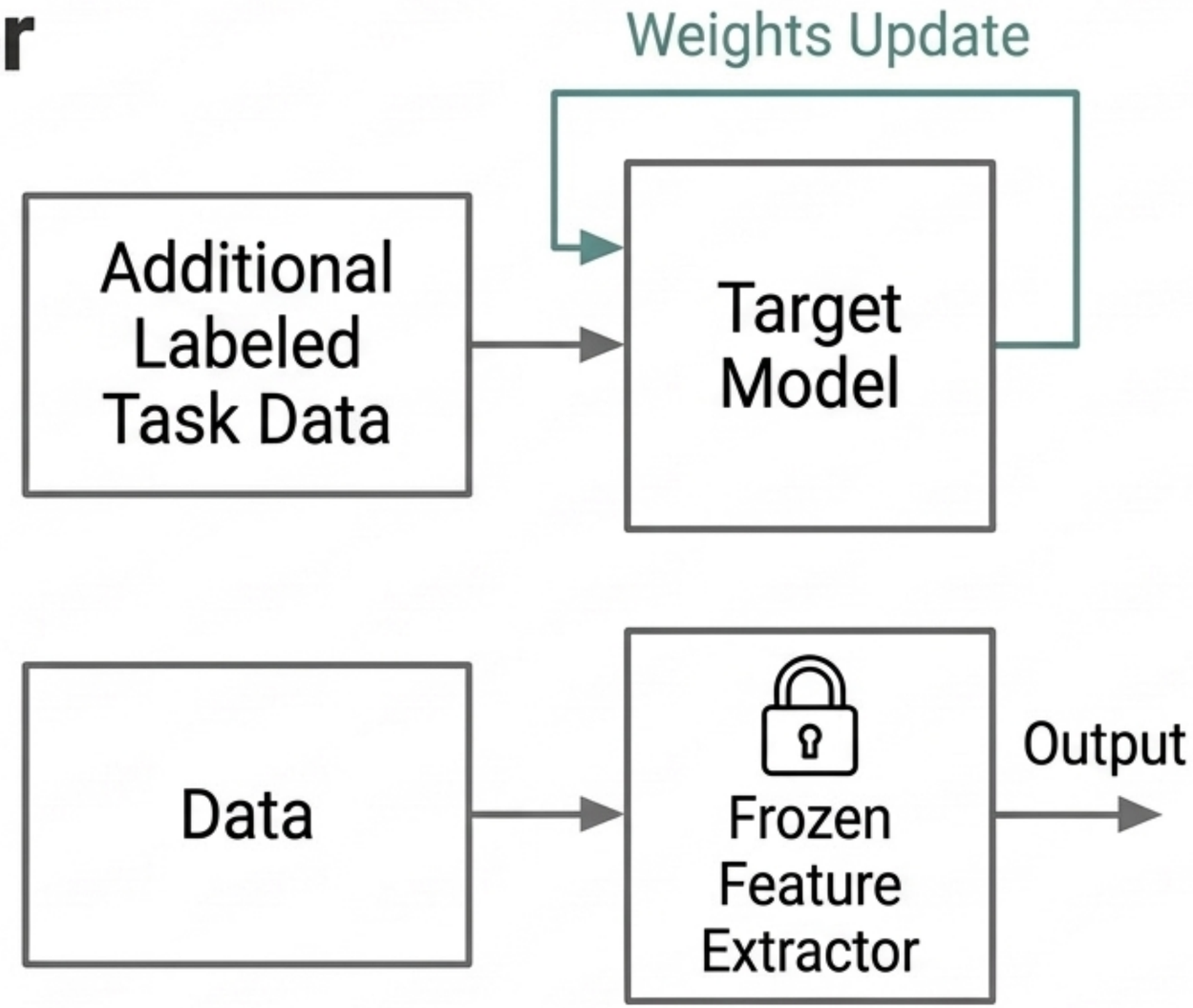
- Initial learning occurs on broad data corpora.
- Process captures broadly useful and reusable language patterns.
- Operation inherently updates the internal model weights.
- Establishes the baseline parameters before downstream engineering use.



Gemini Notebook

Fine-Tuning and Transfer

- Fine-tuning updates parameters using additional task-specific examples.
- Adapts model behavior for specialized, labeled engineering tasks.
- Transfer learning reuses knowledge from earlier training phases.
- Transfer may utilize frozen feature extractors without updating weights.



Gemini Notebook

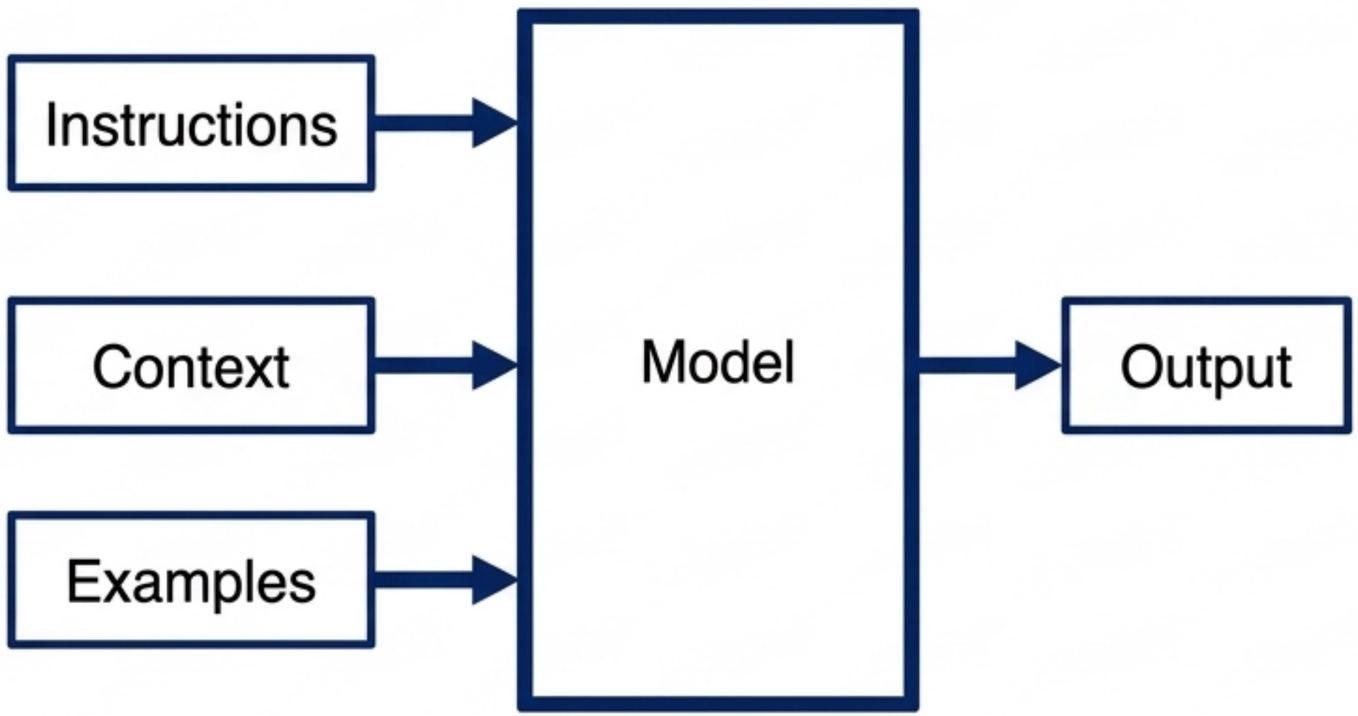
2. Inference Operations

Purpose: Identify operations that provide operational context without altering model parameters.

Gemini Notebook

Prompting Techniques

- Supplies instructions and context exclusively during inference.
- Zero-shot prompting relies on no demonstration examples.
- Few-shot prompting integrates specific demonstration examples.
- Prompting operations do not inherently update model weights.
- Service providers may employ separate data retention policies.



Gemini Notebook

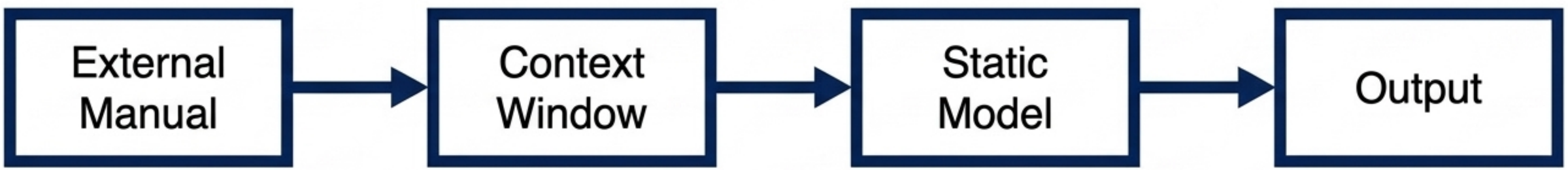
Retrieval Mechanisms

Retrieves external manual passages for in-context application.

Integrates retrieved reference material into the inference prompt.

Retrieval mechanisms do not inherently retrain the model.

Internal parameter weights remain entirely unchanged.



Gemini Notebook

3. Synthesis and Evaluation

Purpose: Consolidate mechanisms and apply engineering evaluation standards.

Gemini Notebook

Mechanism Comparison Matrix

Operation	Typical Role	Weights Change?
Pretraining	Broad corpus learning	Yes
Fine-tuning	Additional task learning	Yes
Prompting	Inference instructions and examples	Not by itself
Retrieval	Source passages in context	Not inherently

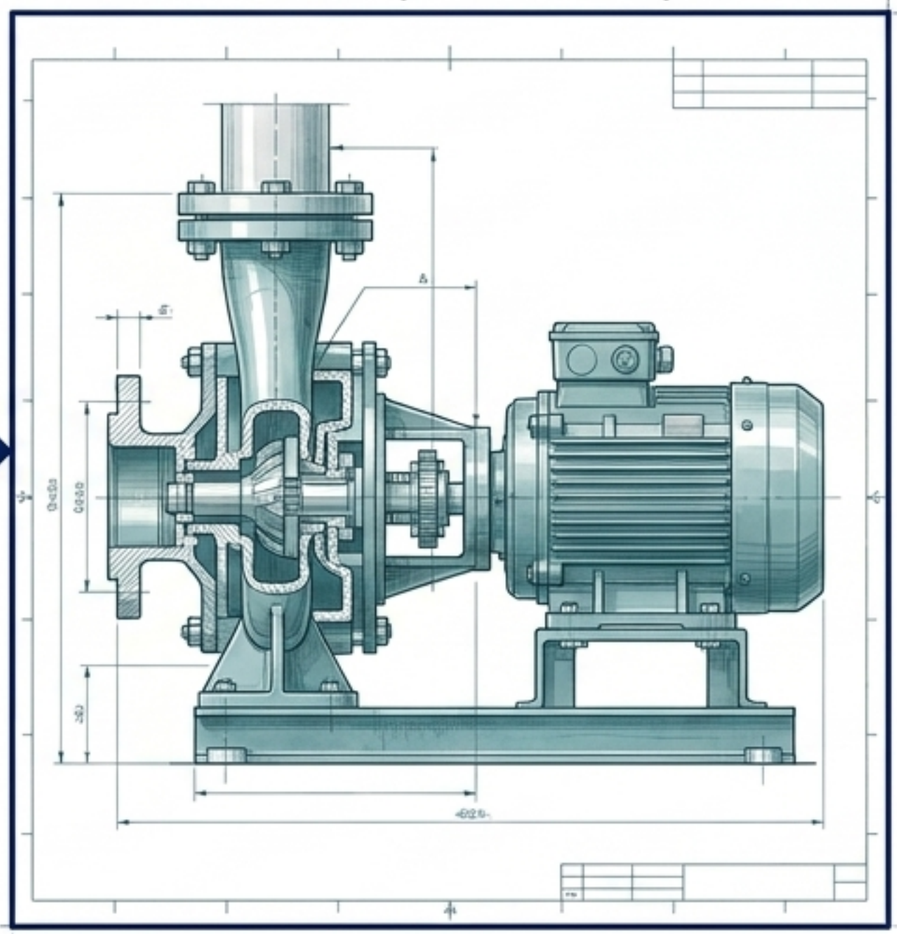
Gemini Notebook

Engineering Evaluation

- Isolate mechanism types when modifying engineering assistants.
- Verify correct source manuals and revision statuses first.
- Always evaluate generated outputs against physical reality.
- Modified weights and optimized context can still yield incorrect claims.

System Outputs

Evaluation Block: Physical Accuracy Check



Gemini Notebook

Key Points

- Pretraining and fine-tuning fundamentally alter internal model parameters.
- Prompting and retrieval provide inference context without changing weights.
- All assistant adaptations necessitate rigorous physical engineering evaluation.

Gemini Notebook